Section 1

Linear Gaussian systems

Inference often combines conditioning and summarizing

Observed y $\rightarrow$ **Condition** $p(z \mid y) = \mathcal{N}\left(z \mid \mu_{z|y}, \Sigma_{z|y}\right)$ $\rightarrow$ **Summarize** $\hat{z} = E[z \mid y] = \mu_{z|y}$

Murphy, Book 2 §2.3.2.2–2.3.2.3

$$E[z \mid y] = \int z p(z \mid y) \, \mathrm{d}z$$

- Conditioning preserves the whole posterior distribution and its uncertainty
- Summarizing extracts a quantity from that posterior; for a Gaussian the mean is $\mu_{z|y}$

A point estimate needs a loss function

Posterior and mean summary Actual slide 63, final state

$$p(z|y) = \mathcal{N}\left(z|\mu_{z|y}, \Sigma_{z|y}\right)$$
$$\hat{z} = E[z|y] = \mu_{z|y}$$

Squared-loss criterion

$$a \in \mathbb{R}^d \quad \text{fixed after observing} \quad y$$
$$E[\|z\|^2|y] < \infty$$
$$L(a, z) = \|z - a\|^2$$
$$R(a|y) = E[L(a, z)|y]$$

Conditional moments and centered residual

$$m = E[z|y], \quad V = \mathrm{Cov}[z|y], \quad r = z - m$$

Earlier Gaussian: $m = \mu_{z|y}$, $V = \Sigma_{z|y}$.

Center at the conditional mean

$$z - a = r + (m - a)$$
$$\|z - a\|^2 = \|r\|^2 + 2r^T(m - a) + \|m - a\|^2$$

- Choosing a reconstruction a requires a criterion: $R(a|y)$ averages its squared error under the posterior.
- Centering separates residual error from action displacement; the cross-term is still present.

The conditional mean minimizes squared-error risk

Pointwise expansion This pair's slide 64, final state

$$m = E[z|y], \quad V = \mathrm{Cov}[z|y], \quad r = z - m$$
$$\|z - a\|^2 = \|r\|^2 + 2r^T(m - a) + \|m - a\|^2$$

Take conditional expectation

$$R(a|y) = E\big[\|r\|^2|y\big] + 2E\big[r^T(m - a)|y\big] + \|m - a\|^2$$

Linearity; m and a are fixed at the observed y.

Cross-term: centering cancels it

$$E[r|y] = E[z|y] - m = 0$$
$$E\big[r^T(m - a)|y\big] = E[r|y]^T(m - a) = 0$$

Residual energy: sum of variances

$$E\big[\|r\|^2|y\big] = \sum_{j=1}^{d} E\big[r_j^2|y\big]$$

$$= \sum_{j=1}^{d} V_{jj} = \mathrm{tr}\, V$$

Centered r: $V = E\big[rr^T|y\big]$.

Substitute the two moment evaluations

$$R(a|y) = E\big[\|z - a\|^2|y\big] = \mathrm{tr}\, V + \|m - a\|^2$$

Minimize the squared-distance term

$\|m - a\|^2 \geq 0$, with equality iff $a = m$

For the earlier Gaussian

$a^*(y) = m = \hat{z} = \mu_{z|y}$

The mean uniquely minimizes squared loss with finite conditional second moments;

$\min_{a \in \mathbb{R}^d} R(a|y) = \mathrm{tr}\, V$

- The covariance trace is the remaining model expected squared error: displacement from the mean adds squared distance.
- Gaussianity is unnecessary.

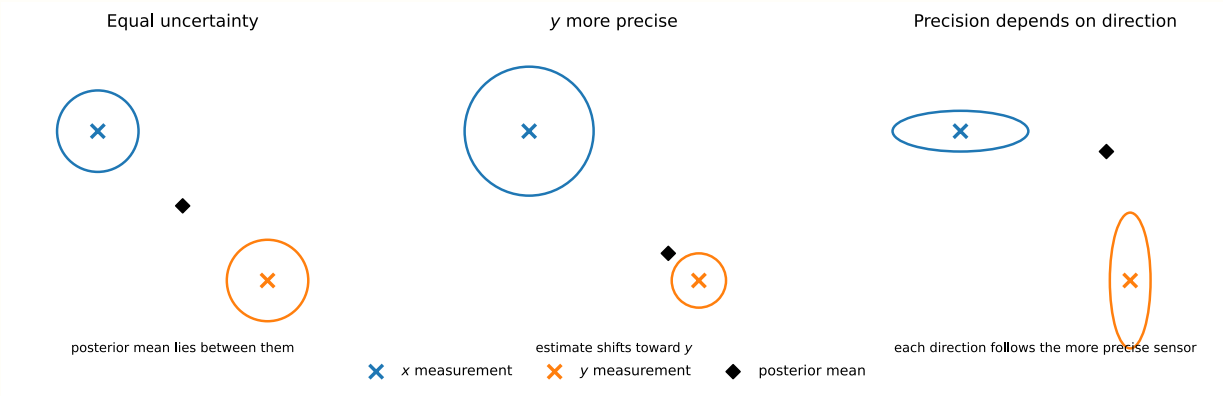
Sensor fusion combines measurements according to their uncertainty

Two noisy measurements of one state

$$z \sim \mathcal{N}(\mu_z, \Sigma_z)\ x \mid z \sim \mathcal{N}(z, \Sigma_x), \quad y \mid z \sim \mathcal{N}(z, \Sigma_y)$$

Same two operations

$$p(z \mid x, y) \quad \rightarrow \quad E[z \mid x, y]$$



Course schematic after Murphy, Book 2 §2.3.2.3, Fig. 2.10; `sensor_fusion_figure.ipynb`

- Equal uncertainty puts the posterior mean between the measurements
- A more precise sensor pulls the posterior mean closer to its measurement
- With anisotropic uncertainty, the weighting can differ by direction